

**BABOLSAR, Iran, Islamic Republic of**WMO: **407360**Lat: **36.6986N**Lon: **52.6425E**Elev: **-69**StdP: **14.73**Time Zone: **3.50 (IRN)**Period: **94-19**WBAN: **99999****Annual Heating, Humidification, and Ventilation Design Conditions**

| Coldest Month | Heating DB |      | Humidification DP/MCDB and HR |      |      |      |      |      | Coldest Month WS/MCDB |      |      |      | MCWS/PCWD to 99.6% DB |      | WSF   |
|---------------|------------|------|-------------------------------|------|------|------|------|------|-----------------------|------|------|------|-----------------------|------|-------|
|               |            |      | 99.6%                         |      |      | 99%  |      |      | 0.4%                  |      | 1%   |      |                       |      |       |
|               | 99.6%      | 99%  | DP                            | HR   | MCDB | DP   | HR   | MCDB | WS                    | MCDB | WS   | MCDB | MCWS                  | PCWD |       |
| (a)           | (b)        | (c)  | (d)                           | (e)  | (f)  | (g)  | (h)  | (i)  | (j)                   | (k)  | (l)  | (m)  | (n)                   | (o)  | (p)   |
| 1             | 35.4       | 37.6 | 30.3                          | 24.4 | 39.0 | 32.9 | 27.3 | 40.8 | 22.6                  | 46.1 | 18.1 | 47.8 | 2.2                   | 90   | 0.361 |

**Annual Cooling, Dehumidification, and Enthalpy Design Conditions**

| Hottest Month | Hottest Month DB Range | Cooling DB/MCWB |      |      |      |      |      | Evaporation WB/MCDB |      |      |      |      |      | MCWS/PCWD to 0.4% DB |      |
|---------------|------------------------|-----------------|------|------|------|------|------|---------------------|------|------|------|------|------|----------------------|------|
|               |                        | 0.4%            |      | 1%   |      | 2%   |      | 0.4%                |      | 1%   |      | 2%   |      |                      |      |
|               |                        | DB              | MCWB | DB   | MCWB | DB   | MCWB | WB                  | MCDB | WB   | MCDB | WB   | MCDB | MCWS                 | PCWD |
| (a)           | (b)                    | (c)             | (d)  | (e)  | (f)  | (g)  | (h)  | (i)                 | (j)  | (k)  | (l)  | (m)  | (n)  | (o)                  | (p)  |
| 8             | 12.8                   | 91.1            | 79.7 | 89.7 | 78.9 | 88.2 | 77.9 | 81.2                | 88.9 | 80.1 | 87.6 | 79.1 | 86.3 | 8.6                  | 330  |

| Dehumidification DP/MCDB and HR |       |      |      |       |      |      |       |      | Enthalpy/MCDB |      |      |      |      |      | Extreme<br>Max WB |
|---------------------------------|-------|------|------|-------|------|------|-------|------|---------------|------|------|------|------|------|-------------------|
| 0.4%                            |       |      | 1%   |       |      | 2%   |       |      | 0.4%          |      | 1%   |      | 2%   |      |                   |
| DP                              | HR    | MCDB | DP   | HR    | MCDB | DP   | HR    | MCDB | Enth          | MCDB | Enth | MCDB | Enth | MCDB |                   |
| (a)                             | (b)   | (c)  | (d)  | (e)   | (f)  | (g)  | (h)   | (i)  | (j)           | (k)  | (l)  | (m)  | (n)  | (o)  |                   |
| 79.0                            | 150.1 | 87.0 | 77.9 | 144.5 | 85.9 | 76.9 | 139.7 | 84.8 | 45.0          | 88.9 | 43.8 | 88.1 | 42.7 | 86.5 | 86.9              |

**Extreme Annual Design Conditions**

| Extreme Annual WS |             |             | Extreme Annual Temperature |             |                    |            | n-Year Return Period Values of Extreme Temperature |             |             |             |             |             |             |              |    |    |
|-------------------|-------------|-------------|----------------------------|-------------|--------------------|------------|----------------------------------------------------|-------------|-------------|-------------|-------------|-------------|-------------|--------------|----|----|
|                   |             |             | Mean                       |             | Standard Deviation |            | n=5 years                                          |             | n=10 years  |             | n=20 years  |             | n=50 years  |              |    |    |
| 1%                | 2.5%        | 5%          | Min                        | Max         | Min                | Max        | Min                                                | Max         | Min         | Max         | Min         | Max         | Min         | Max          | DB | WB |
| (a)               | (b)         | (c)         | (e)                        | (f)         | (g)                | (h)        | (i)                                                | (j)         | (k)         | (l)         | (m)         | (n)         | (o)         | (p)          |    |    |
| (4) <b>17.7</b>   | <b>14.4</b> | <b>12.2</b> | <b>32.1</b>                | <b>94.7</b> | <b>2.7</b>         | <b>2.5</b> | <b>30.2</b>                                        | <b>96.5</b> | <b>28.6</b> | <b>98.0</b> | <b>27.1</b> | <b>99.4</b> | <b>25.2</b> | <b>101.3</b> |    |    |
| (5)               |             |             | <b>31.1</b>                | <b>83.0</b> | <b>2.8</b>         | <b>1.6</b> | <b>29.1</b>                                        | <b>84.1</b> | <b>27.4</b> | <b>85.1</b> | <b>25.8</b> | <b>86.0</b> | <b>23.8</b> | <b>87.2</b>  |    |    |

**Monthly Climatic Design Conditions**

|                                                                                  |             | Annual | Jan   | Feb   | Mar   | Apr   | May   | Jun   | Jul   | Aug   | Sep   | Oct   | Nov   | Dec  |
|----------------------------------------------------------------------------------|-------------|--------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|------|
|                                                                                  |             | (d)    | (e)   | (f)   | (g)   | (h)   | (i)   | (j)   | (k)   | (l)   | (m)   | (n)   | (o)   | (p)  |
| Temperatures,<br>Degree-Days<br>and<br>Degree-Hours                              | DBAvg       | 64.1   | 47.8  | 48.0  | 52.7  | 59.4  | 69.3  | 77.0  | 80.6  | 81.5  | 76.1  | 67.6  | 57.7  | 50.7 |
|                                                                                  | DBStd       | 13.08  | 4.44  | 4.43  | 5.17  | 5.18  | 4.69  | 3.45  | 2.96  | 2.90  | 4.03  | 4.91  | 5.39  | 4.43 |
|                                                                                  | HDD50       | 254    | 94    | 78    | 26    | 2     | 0     | 0     | 0     | 0     | 0     | 0     | 8     | 46   |
|                                                                                  | HDD65       | 2287   | 533   | 475   | 385   | 181   | 15    | 0     | 0     | 0     | 0     | 29    | 225   | 444  |
|                                                                                  | CDD50       | 5409   | 26    | 23    | 109   | 283   | 598   | 811   | 948   | 978   | 783   | 545   | 238   | 67   |
|                                                                                  | CDD65       | 1968   | 0     | 0     | 3     | 12    | 148   | 361   | 483   | 513   | 333   | 110   | 5     | 0    |
|                                                                                  | CDH74       | 17654  | 0     | 0     | 27    | 55    | 538   | 2958  | 5126  | 5853  | 2644  | 440   | 13    | 0    |
|                                                                                  | CDH80       | 5838   | 0     | 0     | 7     | 13    | 65    | 715   | 1880  | 2413  | 694   | 50    | 1     | 0    |
| Wind                                                                             | WSAvg       | 4.4    | 3.7   | 4.4   | 4.9   | 5.0   | 5.3   | 5.3   | 5.1   | 4.8   | 4.3   | 3.7   | 3.3   | 3.3  |
| Precipitation                                                                    | PrecAvg     | 31.1   | 3.2   | 2.4   | 2.3   | 1.4   | 0.9   | 1.1   | 1.2   | 1.2   | 3.6   | 4.6   | 5.6   | 3.8  |
|                                                                                  | PrecMax     | 43.2   | 7.0   | 4.3   | 4.9   | 3.3   | 2.8   | 5.2   | 4.1   | 4.1   | 6.7   | 10.7  | 9.7   | 7.3  |
|                                                                                  | PrecMin     | 17.9   | 1.1   | 0.4   | 0.5   | 0.0   | 0.1   | 0.0   | 0.1   | 0.0   | 0.2   | 0.9   | 0.6   | 0.5  |
|                                                                                  | PrecStd     | 5.9    | 1.4   | 1.2   | 1.1   | 0.9   | 0.6   | 1.1   | 1.1   | 1.1   | 1.7   | 2.3   | 2.1   | 1.8  |
| Monthly Design<br>Dry Bulb<br>and<br>Mean Coincident<br>Wet Bulb<br>Temperatures | 0.4%        | DB     | 65.4  | 66.5  | 77.4  | 80.3  | 85.7  | 89.6  | 92.0  | 93.0  | 90.0  | 84.3  | 75.6  | 67.8 |
|                                                                                  |             | MCWB   | 53.8  | 55.2  | 61.2  | 65.6  | 72.1  | 77.7  | 80.0  | 81.0  | 78.6  | 72.6  | 67.4  | 57.9 |
|                                                                                  | 2%          | DB     | 59.9  | 60.5  | 69.1  | 73.5  | 81.2  | 87.6  | 90.7  | 91.4  | 87.8  | 80.6  | 71.2  | 62.8 |
|                                                                                  |             | MCWB   | 52.9  | 52.7  | 58.5  | 63.4  | 70.4  | 76.2  | 79.4  | 80.3  | 77.6  | 72.4  | 64.4  | 56.6 |
|                                                                                  | 5%          | DB     | 56.7  | 56.6  | 63.3  | 70.0  | 79.0  | 85.9  | 89.2  | 90.0  | 85.8  | 78.4  | 68.3  | 60.1 |
|                                                                                  |             | MCWB   | 51.6  | 51.1  | 56.1  | 62.2  | 69.9  | 75.3  | 78.5  | 79.4  | 76.5  | 70.9  | 62.4  | 55.2 |
|                                                                                  | 10%         | DB     | 54.5  | 54.1  | 59.2  | 66.8  | 77.0  | 84.2  | 87.6  | 89.1  | 84.0  | 76.2  | 66.2  | 57.8 |
|                                                                                  |             | MCWB   | 50.4  | 49.9  | 54.1  | 60.4  | 69.0  | 74.2  | 77.5  | 78.9  | 75.2  | 69.3  | 61.0  | 53.7 |
| Monthly Design<br>Wet Bulb<br>and<br>Mean Coincident<br>Dry Bulb<br>Temperatures | 0.4%        | WB     | 56.0  | 56.2  | 62.9  | 67.0  | 74.3  | 79.2  | 82.3  | 82.8  | 80.6  | 75.3  | 68.4  | 60.0 |
|                                                                                  |             | MCDB   | 61.3  | 64.4  | 74.6  | 76.6  | 81.5  | 86.9  | 90.2  | 90.5  | 88.1  | 80.5  | 73.4  | 64.4 |
|                                                                                  | 2%          | WB     | 53.9  | 53.4  | 58.9  | 64.6  | 72.4  | 77.7  | 80.8  | 81.6  | 78.7  | 73.3  | 65.3  | 57.6 |
|                                                                                  |             | MCDB   | 58.5  | 58.3  | 66.3  | 72.0  | 79.1  | 85.1  | 88.5  | 89.3  | 85.7  | 79.0  | 69.6  | 61.6 |
|                                                                                  | 5%          | WB     | 52.4  | 51.8  | 56.4  | 62.7  | 71.0  | 76.5  | 79.7  | 80.7  | 77.3  | 71.7  | 63.5  | 55.7 |
|                                                                                  |             | MCDB   | 56.1  | 55.5  | 62.8  | 68.7  | 77.3  | 83.6  | 87.1  | 88.3  | 84.1  | 77.0  | 67.7  | 59.2 |
|                                                                                  | 10%         | WB     | 51.0  | 50.7  | 54.4  | 61.0  | 69.6  | 75.4  | 78.6  | 79.8  | 76.1  | 70.1  | 61.6  | 54.3 |
|                                                                                  |             | MCDB   | 54.1  | 53.7  | 59.0  | 66.1  | 75.5  | 82.3  | 85.8  | 87.1  | 82.5  | 75.1  | 65.3  | 57.4 |
| Mean Daily<br>Temperature<br>Range                                               | 5% DB       | MDBR   | 11.8  | 10.4  | 9.9   | 9.9   | 10.2  | 11.1  | 11.8  | 12.8  | 12.5  | 13.3  | 12.9  | 12.3 |
|                                                                                  |             | MCDBR  | 15.9  | 15.7  | 18.7  | 16.1  | 13.6  | 13.0  | 13.5  | 14.4  | 14.2  | 15.4  | 15.7  | 15.9 |
|                                                                                  |             | MCWBR  | 10.4  | 10.0  | 10.2  | 8.0   | 6.6   | 5.7   | 6.0   | 7.1   | 7.1   | 8.9   | 10.0  | 10.9 |
|                                                                                  | 5% WB       | MCDBR  | 14.3  | 13.4  | 17.1  | 14.4  | 12.2  | 11.9  | 12.6  | 13.6  | 13.3  | 14.3  | 14.3  | 14.1 |
|                                                                                  | MCWBR       | 9.9    | 9.1   | 9.6   | 7.7   | 6.5   | 5.9   | 6.1   | 7.3   | 7.2   | 9.1   | 9.7   | 10.3  |      |
| Clear-Sky<br>Solar<br>Irradiance                                                 | taub        | 0.401  | 0.445 | 0.502 | 0.575 | 0.581 | 0.589 | 0.637 | 0.601 | 0.576 | 0.530 | 0.455 | 0.408 |      |
|                                                                                  | taud        | 2.178  | 2.060 | 1.941 | 1.821 | 1.855 | 1.868 | 1.788 | 1.857 | 1.889 | 1.955 | 2.082 | 2.159 |      |
|                                                                                  | Ebn at Noon | 241    | 244   | 241   | 231   | 233   | 230   | 218   | 222   | 218   | 214   | 218   | 227   |      |
|                                                                                  | Edh at Noon | 36     | 45    | 56    | 66    | 65    | 64    | 69    | 63    | 58    | 50    | 39    | 34    |      |
| All-Sky Solar<br>Radiation                                                       | RadAvg      | 731    | 913   | 1197  | 1508  | 1853  | 2047  | 1940  | 1836  | 1473  | 1111  | 791   | 646   |      |
|                                                                                  | RadStd      | 64     | 102   | 102   | 169   | 101   | 127   | 154   | 198   | 91    | 82    | 68    | 65    |      |